

ENERGY CHANGES – REACTION PROFILES

KEY VOCABULARY

Reaction profile: a graph of energy against progress of reaction.

Activation energy (E_a): the minimum energy needed to start a reaction.

Energy change (ΔH): the difference between reactant and product energy.

Catalyst: lowers the activation energy.

COMMON MISCONCEPTIONS

~~Activation energy is the same as ΔH .~~

✓ E_a is the hump; ΔH is the start-to-end change.

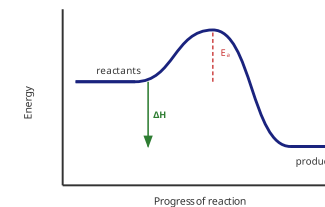
~~A catalyst changes ΔH .~~

✓ A catalyst lowers E_a but ΔH is unchanged.

~~Exothermic profiles go uphill.~~

✓ Exothermic products end LOWER than reactants.

EXOTHERMIC PROFILE



The peak shows E_a ; the drop shows energy **released**.

1

LABEL THE PROFILE

On a profile, state what the **peak** and the **overall drop** represent. [2 marks]

2

DEFINE E_a

State what is meant by **activation energy**. [1 mark]

3

EXO OR ENDO

Explain how a profile shows that a reaction is exothermic. [2 marks]

4

CATALYST EFFECT

Explain how a catalyst changes the profile. [2 marks]

5

FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

A catalyst provides a route with a _____ energy.

activation

lower

raise

overall

6

SKETCH

Sketch the profile of an endothermic reaction with its E_a . [3 marks]
